

Quiz (Carolina Reaper)

- Q1.** A shipping company has $12x + 18$ crates to transport. What is the greatest common factor of this expression?
(A) $2(6x + 9)$
(B) $3(4x + 6)$
(C) $6(2x + 3)$
(D) $12(x + 3)$
- Q2.** The traffic flow on a highway is modeled by the expression $20x^2 + 30x$. What is the factored form of this expression?
(A) $10x(2x + 3)$
(B) $5x(4x + 6)$
(C) $10x(2x + 1)$
(D) $5(4x^2 + 6x)$
(E) $2(10x^2 + 15x)$
- Q3.** A delivery truck travels $8x + 12$ miles per hour. What is the greatest common factor of this expression?
(A) $2(4x + 6)$
(B) $4(2x + 3)$
(C) $8(x + 3)$
(D) $4x + 12$
(E) $8x + 12$
- Q4.** The cost of shipping a package is modeled by the expression $15x + 25$. What is the factored form of this expression?
(A) $5(3x + 5)$
(B) $3(5x + 25)$
(C) $15x + 5$
(D) $25x + 15$
- Q5.** A logistics company has $24x + 36$ containers to transport.
- What is the greatest common factor of this expression?
(A) $12(2x + 3)$
(B) $6(4x + 6)$
(C) $12(2x + 1)$
(D) $24(x + 3)$
- Q6.** The time it takes for a package to be delivered is modeled by the expression $9x + 12$. What is the factored form of this expression?
(A) $3(3x + 4)$
(B) $3x + 12$
(C) $9x + 3$
(D) $12x + 9$
(E) $3(3x + 2)$
- Q7.** A transportation company has $16x + 20$ vehicles to maintain. What is the greatest common factor of this expression?
(A) $4(4x + 5)$
(B) $8(2x + 5)$
(C) $16x + 4$
(D) $20x + 16$
- Q8.** The volume of cargo that can be transported by a ship is modeled by the expression $12x^2 + 15x$. What is the factored form of this expression?
(A) $3x(4x + 5)$
(B) $3(4x^2 + 5x)$
(C) $3x(4x + 3)$
(D) $3(4x^2 + 3x)$
(E) $12x(x + 5)$

Open Ended Questions (FRQ)

- Q9.** A logistics company has $25x + 30$ crates to transport. Factor out the greatest common factor and simplify the expression. Show your work.
- Q10.** The cost of shipping a package is modeled by the expression $18x + 24$. Factor out the greatest common factor and simplify the expression. Show your work and explain your reasoning.

Chef's Tips

- Read each question carefully before answering.
- Show all work and explain your reasoning for free-response questions.
- Use the factoring method to simplify expressions whenever possible.